Baixin Xu

School of Computer Science and Engineering
Nanyang Technological University
50 Nanyang Avenue, Singapore 639798

(+65) 90522828
baixin001@e.ntu.edu.sg
xubaixinxbx.github.io

EDUCATION

Ph.D. Nanyang Technological University, supervised by *Prof. Ying He*, 08/2023 - now

B.E. Huazhong University of Science and Technology, 09/2018 - 07/2022

RESEARCH PROFILE

Ph.D. student in computer vision and computer graphics, seeking research engineer/scientist roles in 3D vision, generative AI, neural rendering, and world models. My work focuses on reconstructing, editing, and generating high-quality 3D geometry and interactive visual worlds from sparse or multimodal inputs.

PUBLICATIONS

* *denotes equal contribution. Publications are grouped by research direction to make the technical narrative clearer for industry research roles.*

3D Reconstruction, Neural Rendering, and Geometry

Baixin Xu, Jiarui Zhang, Kwan-Yee Lin, Chen Qian, Ying He. "Deformable Model Driven Neural Rendering for High-Fidelity 3D Reconstruction of Human Heads Under Low-View Settings." **ICCV 2023**

Baixin Xu, Jiangbei Hu, Fei Hou, Kwan-Yee Lin, Wayne Wu, Chen Qian, Ying He. "Parameterization-driven Neural Surface Reconstruction for Object-oriented Editing in Neural Rendering." **ECCV 2024**

Baixin Xu, Jiangbei Hu, Jiaze Li, Ying He. "GSurf: 3D Reconstruction via Signed Distance Fields with Direct Gaussian Supervision." *arXiv 2024*

Qingyuan Zhou, Yuehu Gong, Weidong Yang, Jiaze Li, Yeqi Luo, **Baixin Xu**, Shuhao Li, Ben Fei, Ying He. "MGSR: 2D/3D Mutual-boosted Gaussian Splatting for High-fidelity Surface Reconstruction under Various Light Conditions." **ICCV 2025**

Daisheng Jin, Jiangbei Hu, **Baixin Xu**, Yuxin Dai, Chen Qian, Ying He. "SFDM: Robust Decomposition of Geometry and Reflectance for Realistic Face Rendering from Sparse-view Images." **CVPR 2025**

Jiayi Kong, Xurui Song, Huai Shuo, **Baixin Xu**, Jun Luo, Ying He. "Do Not DeepFake Me: Privacy-Preserving Neural 3D Head Reconstruction Without Sensitive Images." **AAAI 2025**

Jiangbei Hu, Haobo Wang, **Baixin Xu**, Nan Ding, Zhimao Lu, Na Lei, Ying He. "AquaSplatting: A Hybrid 3D Representation for Robust Underwater Scene Reconstruction via Dual-Branch Rendering." **AAAI 2026**

Jiangbei Hu, Ben Fei, **Baixin Xu**, Fei Hou, Shengfa Wang, Na Lei, Weidong Yang, Chen Qian, Ying He. "TopoGen: Topology-Aware 3D Generation with Persistence Points." **Pacific Graphics / Computer Graphics Forum 2025**

Jiangbei Hu, Ying He, **Baixin Xu**, Shengfa Wang, Na Lei, Zhongxuan Luo. "IF-TONIR: Iteration-Free Topology Optimization Based on Implicit Neural Representations." *Computer-Aided Design 2024*

World Models and Multimodal Generation

Zile Wang*, Zexiang Liu*, Jiaxing Li*, Kaichen Huang*, **Baixin Xu***, ..., Yangguang Li, Yahui Zhou "Matrix-Game 3.0: Real-Time and Streaming Interactive World Model with Long-Horizon Memory." *Tech Report 2026*

Xianglong He*, Chunli Peng*, Zexiang Liu*, Boyang Wang*, *et al.*, **Baixin Xu**, *et al.* "Matrix-Game 2.0: An Open-Source Real-Time and Streaming Interactive World Model." *Tech Report 2025*

Zhongqi Yang, Wenhang Ge, Yuqi Li, Jiaqi Chen, Haoyuan Li, Mengyin An, Fei Kang, Hua Xue, **Baixin Xu**, *et al.* "Matrix-3D: Omnidirectional Explorable 3D World Generation." *Tech Report 2025*

Hongyang Wei*, **Baixin Xu***, Hongbo Liu*, Size Wu, ..., Yangguang Li, Yahui Zhou "Skywork UniPic 2.0: Building Kontext Model with Online RL for Unified Multimodal Model." *Tech Report 2025*

Hongyang Wei, Hongbo Liu, Zidong Wang, Yi Peng, **Baixin Xu**, *et al.* "Skywork UniPic 3.0: Unified Multi-Image Composition via Sequence Modeling." *Tech Report 2026*

EXPERIENCE

Skywork AI / Kunlun 2050, Singapore, Research Engineer (IPP), 10/2024 - 03/2026. Worked on interactive world models, unified multimodal generation, and large-scale generation systems; contributed to Matrix-Game and Skywork UniPic projects.

S-Lab, NTU, Singapore, Research Assistant, 11/2022 - 10/2024. Advised by *Prof. Ying He*; worked on neural surface reconstruction, Gaussian/SDF representations, and editable neural rendering.

Shanghai AI Laboratory, China, Research Intern, 09/2022 - 06/2023. Advised by *Dr. Kwan-Yee Lin*; worked on high-fidelity neural head reconstruction under sparse-view settings.

SenseTime Research, China, Research Intern, 07/2022 - 09/2022. Worked on neural rendering and 3D head reconstruction.

SERVICES

Reviewer: CVPR 2024/2025, ECCV 2024, ICCV 2025, NeurIPS 2025, ACM MM 2024, AAAI 2025/2026, ICLR 2025, Pacific Graphics 2023/2024, CVM-J, TVCG, TIP.

AWARDS AND HONORS

2022 Outstanding graduates of HUST.

TECHNICAL SKILLS

Python, PyTorch, Linux, Git; 3D Gaussian Splatting, neural implicit representations, signed distance fields, neural rendering, 3D reconstruction, 3D generation, image/video generation.

Updated June 2026